BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (11TH CLASS) EXAMINATION, 2023

PHYSICS

Past Paper 2023 • Paper Code: 6475

Retyped faithfully from the original board paper — same questions, no changes.

Physics (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 6475

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill that circle in front of that question number with marker or pen. Cutting or filling two or more circles will result in zero mark in that question.

Q1. How many number of zeros in 7.4000 and 8000 kg are significant, if the later quantity has 1 kg least count?

(A) None and none
(B) Three and three
(C) Three and two
(D) Two and one

Q2. What choice would you have to pick for the percentage uncertainty in measuring 2.3cm with meter rod and 2.45cm with vernier caliper?

(A) 4.3% and 0.4%
(B) 4.0% and 0.1%
(C) 0.1% and 0.01%
(D) 3.9% and 0.52%

Q3. The distance of point P of position vector r = 3i + 3j + 3k from the origin is:

(A) 3.0 unit
(B) 6.4 unit
(C) 5.2 unit
(D) 2.5 unit

Q4. Two vectors to be combined have magnitudes 60N and 35N with different directions. Pick the correct answer:

(A) 95 N
(B) 25 N
(C) Both A & B
(D) Can be any value between option A & B

Q5. What is the distance travelled by a body in the following velocity time graph?

(A) 100 m
(B) 20 m
(C) 10 m
(D) 5 m

Q6. Water flows out from a pipe at 3Kgs-1 and its velocity changes from 5ms-1 to zero on striking the ball, then force is:

(A) 10 N
(B) 15 N
(C) 20 N
(D) 25 N

Q7. Solar energy at normal incidence outside the Earth's atmosphere is:

(A) 1.0 Wm-2
(B) 1.4 Wm-2
(C) 1.0k Wm-2
(D) 1.4k Wm-2

Q8. For one radian, arc length 'S' and radius 'r' of circle has the relation:

(A) S > r
(B) r > S
(C) S = r
(D) S = 2r

Q9. Centripetal force performs:

(A) Positive work
(B) No work
(C) Negative work
(D) Maximum work

Q10. Which one is correct?

(A) 1 torr=13.33 Nm-2
(B) 2 torr=26.66 Nm-2
(C) 5 torr=65.65 Nm-2
(D) 10 torr=1333 Nm-2

Q11. Which other quantity has the same unit as spring constant has Nm-1?

(A) Torque
(B) Momentum
(C) Surface tension
(D) Rotational K.E

Q12. If a wave is travelling in a rarer medium and incident on a denser medium then the phase change is:

(A) 360 degrees
(B) 180 degrees
(C) 90 degrees
(D) 0 degrees

Q13. When air column is closed from one end then the fundamental frequency becomes:

(A) 1.0(v/l)
(B) 0.5(v/l)
(C) 0.75(v/l)
(D) 0.25(v/l)

Q14. When fringe spacing in Young's double slit experiment is increased by increasing wavelength then number of fringes will be:

(A) Decreased
(B) Increased
(C) Constant
(D) Disappear

Q15. For incident angles equal to or greater than the critical angle, the glass-air boundary will act as:

(A) Lens (Biconvex)
(B) Mirror (Plane)
(C) Lens (Concave)
(D) Lens (Plano convex)

Q16. When the work is done at the cost of internal energy, then the equation becomes:

(A) W = Delta U

(B) W = Q

(C) W = -Delta U

(D) W = Delta V

Q17. Pick the correct condition for the relation Cp - Cv = R:

(A) Internal energy is kept constant for both processes

(B) Pressure is kept constant for both processes

(C) Temperature is kept constant for both processes

(D) Volume is kept constant for both processes

Physics (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68

Note: SECTION - I

Q2. Write short answers to any EIGHT parts.

i. The period of simple pendulum is measured by a stop watch. What type of errors are possible in the time period?

ii. Write the dimensions of (a) pressure (b) density.

iii. Given that V = (5.2 +/- 0.1) volt. Find its percentage uncertainty.

iv. What are supplementary units? Define only one unit.

v. Under what circumstances would a vector have components that are equal in magnitude?

vi. Suppose the sides of a closed polygon represent vector arranged head to tail rule. What is the sum of these vectors?

vii. Define the two conditions of equilibrium.

viii. Explain the circumstances in which the velocity v and acceleration a of a car are (a) anti-parallel (b) v is zero but a is not zero.

ix. Define impulse and show that how it is related to linear momentum?

x. What is isolated system? State the law of conservation of momentum.

xi. What is the effect on the speed of a fighter plane chasing another when it opens the fire?

xii. Explain the difference between laminar flow and turbulent flow.

Q3. Write short answer to any EIGHT parts.

i. Calculate the work done in kilo joules in lifting a mass of 10kg (at steady velocity) through a vertical height of 10cm.

ii. What sort of energy is in the (a) compressed spring (b) water in high dam?

iii. Define escape velocity. Wirte the formula to find escape velocity.

iv. Why does a diver change his body positions before and after diving in the pool?

v. When mud flies off the tyre of a moving bicycle, in what direction does it fly? Explain.

vi. Show that S = r theta

vii. What is the total distance travelled by an object moving with simple harmonic motion in time equal to its period, its amplitude is A?

viii. Explain the relation between total energy, potential energy and kinetic energy of a body oscillating SHM.

ix. Draw the graph between amplitude and time in damped oscillations.

x. Explain the terms crest, trough, node and antinode.

xi. Explain why sound travels faster in warm air than in cold air.

xii. Speed of sound in air at 0 degrees C is 332 ms-1. Find its speed at 15 degrees C.

Q4. Write short answer to any SIX parts.

i. How is the distance between interference fringes affected by the separation between the slits of Young's experiment? Can fringes disappear?

ii. An oil film spreading over a wet footpath shows colours. Explain how does it happen?

iii. Find the grating element of the diffraction grating containing 2000 lines / cm.

iv. Explain briefly the single mode step index fiber.

v. Why would it be advantageous to use blue light with a compound microscope?

vi. Give at least two postulates of kinetic theory of gases.

vii. Derive Boyle's law on the basis of kinetic theory of gases.

viii. Give an example of a process in which no heat is transferred to or from the system but temperature of the system changes?

ix. Is it possible to construct a heat engine that will not expel heat into the atmosphere? Explain.

> Note: SECTION - II

> Note: Attempt any THREE questions. Each question carries 08 Marks.

Q5.

(a) What is a scalar product? Discuss its physical interpretation and write its three characteristics.

(b) Ten bricks, each 6.0cm thick and mass 1.5kg lie flat on a table. How much work is required to stack them one on the top of another?

Q6.

(a) State and explain law of conservation of linear momentum.

(b) A gramophone record turntable accelerates from rest to an angular velocity of 45.0 rev min-1 in 1.60s.

Q7.

(a) Define Stoke's law and show that the terminal velocity is directly proportional to square of radius of the water droplet.

(b) A heat engine performs 100J of work and at the same time rejects 400J of heat energy to the cold reservoirs. What is the efficiency of the engine?

Q8.

(a) Discuss the motion of a horizontal mass spring system and find the values of time period, instantaneous displacement and instantaneous velocity.

(b) A pipe has a length of 1m. Determine the frequencies of the fundamental and the first two harmonics if the pipe is closed at one end.

Q9.

(a) Explain diffraction of x-rays by crystals and derive Bragg's equation.

(b) A simple astronomical telescope in its normal adjustment has an objective of focal length 100cm and an eye piece of focal length 5.0cm (i) where is the final image formed (ii) calculate the angular magnification.